

Sentiment Analysis of Religious Tweets

Pulkit Gigoo

Electronics Engineering
Vishwakarma Institute of Technology
Pune, India
pulkit.gigoo18@vit.edu

Rishabh Tiwari

Electronics Engineering
Vishwakarma Institute of Technology
Pune, India
rishabh.tiwari18@vit.edu

Saikrishna Ramesh

Electronics Engineering
Vishwakarma Institute of Technology
Pune, India
saikrishna.ramesh18@vit.edu

Sanket Bendrey

Electronics Engineering
Vishwakarma Institute of Technology
Pune, India
sanket.bendrey18@vit.edu

Prof. Rupali Tornekar

Electronics and Communications Dept.
Vishwakarma Institute of Technology
Pune, India
rupali.tornekar@vit.edu

Abstract—Sentiment analysis or opinion mining is an important type of text analysis that aims to support decision making by extracting and analyzing opinion-oriented text, identifying positive and negative opinions, and measuring how positively or negatively an entity (i.e., people, organization, event, location, product, topic, etc.) is regarded. As more and more users express their political and religious views on Twitter, tweets become valuable sources of people's opinions. This paper proposes a comprehensive guide of using Python to extract, preprocess, and analyze tweets about religion. The analysis is about implementing Topic Modelling (LDA), Sentiment Analysis (Gensim), and Hate Speech Detection (Hate Sonar) models.

Keywords—Sentiment Analysis; Tweets; Text Analysis; Religion

I. INTRODUCTION

On Web 2.0, user-generated content, which is material submitted by users who interact with social network sites, is a major theme. Twitter is a social networking and microblogging service where users send messages (a.k.a., tweets) to a network of associates from a variety of devices. A tweet is a text-based post and only has 140 characters, which is approximately the length of a typical newspaper headline and subhead. As more and more users post reviews about products and services they use, or express their political and religious views on Twitter, tweets become valuable sources of people's opinions and sentiments. Given its popularity, Twitter is seen as a potential new form of eWOM (electronic word-of mouth) marketing by the businesses and organizations concerned with reputation management.

Sentiment analysis (or opinion mining) is stated as “the computational study of opinions, sentiments and emotions expressed in text”. Reviews tend to be longer and more verbose than tweets which may only be a few words long and often contain significant spelling errors. In this study, we focus on the tweets sentiment analysis that is to automatically identify whether a piece of text expresses a positive or negative opinion using LDA, Gensim, Vader and Hate Speech Detection models.

II. LITERATURE SURVEY

The first paper we researched was paper published by **Hurmet Noka** on **Text mining tweets on religion**. This paper presents a comprehensive guide of using Python to extract, pre-process, and analyze tweets about religion. The analysis is about implementing **Topic Modeling (LDA)**, **Sentiment Analysis (Gensim)**, and **Hate Speech Detection (HateSonar)**

models. The second paper we looked at was on **Sentiment Analysis on tweets about Religion**. The authors are **Xujuan Zhou, Xiaohui Tao, Jianming Yong and Zhenyu Yang**. They have focused on tweet sentiment analysis and automatically identify whether the tweet is positive, negative etc. using Tweets Sentiment Analysis Model (TSAM). The third paper we researched was on **Analysis of Inter-Religious tolerance sentiments in Indonesia on conversations on social media twitter** and authors are **Yogie Pribadi, Noor Hafidz, Yamin Nuryamin, Windu Gata**. They compared the issues of tolerance between religious communities in Indonesia by using SVM, Naïve-Bayes algorithm, Decision Tree and K-Nearest Neighbor (KNN). The last paper we researched was **Identification of Hatred Speeches on Twitter** by **Tehseen Zia, Basit Shahzad and M.Shehbaz Akram** which focused on filtering tweets/messages over Twitter with hatred speeches using SVM, KNN and Naïve Bayes.

III. METHODOLOGY

A. Data Extraction and Pre-Processing

To get the tweets, we use a public python script, which enables capturing old tweets, thus bypassing the limitation of the 7-days period of Twitter API. All you need to do is adjust the searching filters and run the program. For our study, we

extract tweets containing the phrase “*religion is.*” To reduce the bias of certain isolated events affecting the feeling towards religion (i.e., Charlie Hebdo attacks), we extend the timeframe to about five years. We extracted 1000 tweets/month starting from January 2015 until October 2019. This results in about 57'351 tweets that are then loaded into a dataframe, ready for pre-processing.”.

Below is the whole analysis process in a schematic format

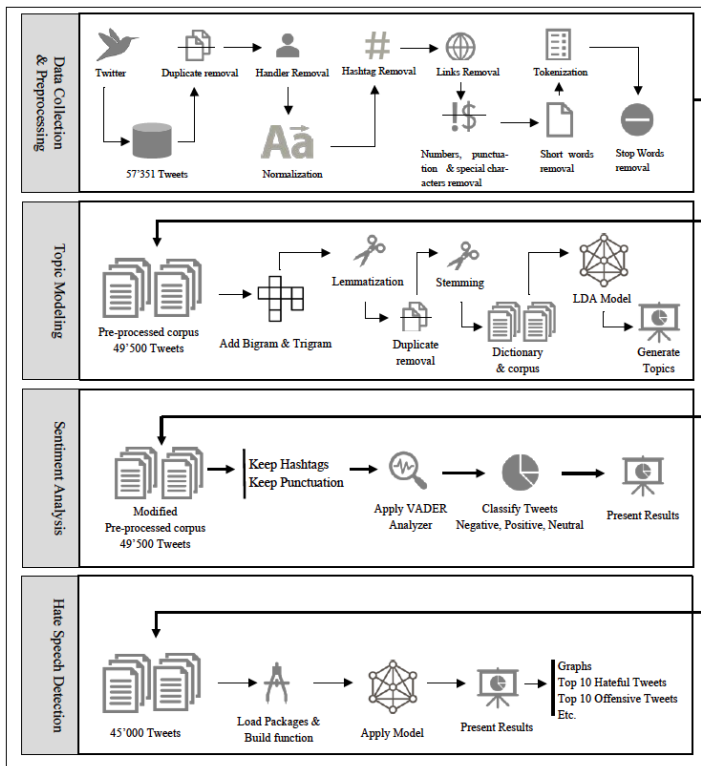


Fig1. Analysis Steps

The preprocessing phase effects are illustrated below, where the length of cleaned tweets is reduced considerably. As shown, most tweets have less than ten tokens after preprocessing, unlike in original tweets where most tweets have about 20 words. On the second set of graphs, the effects are more visible as the tweet length went from about 150 for most of the tweets to about 50 characters after the cleaning stage. This phase is critical as it reduces the dimensionality and results in significantly valuable tokens for the models, which is explained in the next sections.

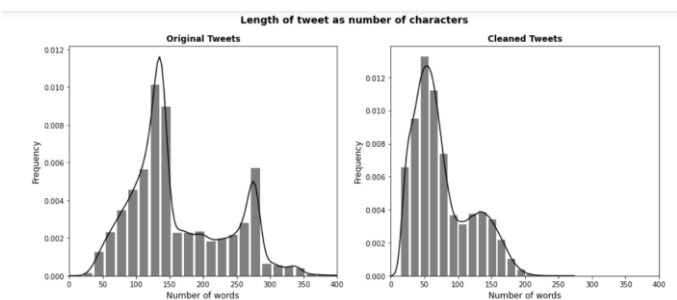


Fig2. Length of tweet as number of characters

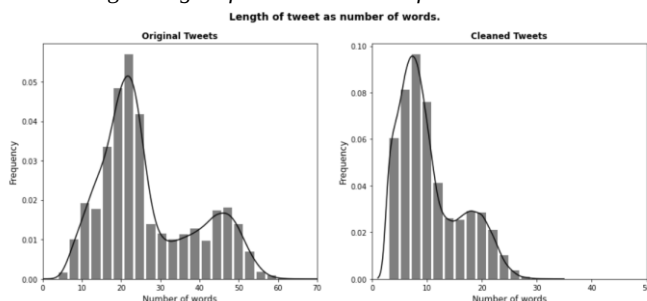


Fig2. Length of tweet as number of words

B. Implementing different models

We have implemented the following models:

1.Linear Discriminant Analysis (LDA)

The core idea of the LDA algorithm can be understood as a generative process where documents are defined by a probability distribution over a set of topics T , and a probability distribution of discrete words, in turn, establishes each topic. Having said that, we load the pre-processed data: After data is loaded, we proceed by adding **Bigrams** and **Trigrams**. A sequence of words which often occurs together when expressing a particular meaning. A sequence of N words is known as N -Grams, as theoretically N can be of any length; the most common are pairs of words (Bi-grams) and a sequence of three words (Tri-grams). The next step is *lemmatization*, an essential step for many applications of text mining. Lemmatization takes into consideration the context and converts the word to its base form; for instance, the term “*hugging*” is converted to “*hug*” and “*best*” to “*good*.” For the lemmatization task, the package used is *spaCy*, an open-source library with many pre-built models for natural language processing.

After this step, we dropped duplicates once more, as short tweets composed of very few tokens when lemmatized may lead to duplicate rows.

Another useful technique is *word stemming*, which is the process of transforming a word into its root form. Unlike lemmatization, which was mentioned earlier, stemming is a more aggressive approach as suffix cuttings often result in non-meaningful English words. For instance, the word “*animals*” would be lemmatized as “*animal*,” but the Porter Stemmer yields “*anim*.” We decided to implement both to help in dimensionality reduction.

Before building the LDA model, we have to create two main inputs: the dictionary and the corpus, which are created using functions from the *Gensim* package.

Gensim assigns a unique Id to each word, and then the corpus is represented as a tuple (*word_id*, *word_frequency*).

BUILD LDA MODEL

Now it is the moment that we initialize with number of topics $k=10$, which then will be subject of adjustments. After this step, we may directly generate the topics or search for the optimal model, using **coherence score** as a measure for each model having a different number of topics.

Below is given the distribution of tweets among topics, we can notice that the first three topics are more dominant:

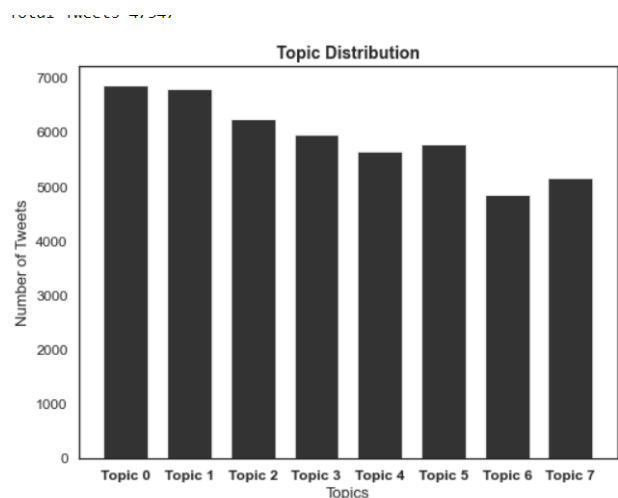


Fig3. Topic Distribution

2.Valence Aware Dictionary for sentiment Reasoning (VADER)

VADER stands for Valence Aware Dictionary for sentiment Reasoning and was developed as a rule-based model for sentiment analysis. It takes into consideration punctuation, capitalization, degree modifiers, conjunctions, Tri-gram preceding when assigning sentiment values: *negative*, *positive*, *neutral*. These features make VADER sentiment analyzer achieve remarkable results when classifying social media texts like tweets and make it a suitable tool to conduct our analysis. The pre-processing phase is slightly different from the one employed for Topic Modelling because of the features that VADER sentiment analyzer emodies, the steps taken are listed below, whereas the code is the same as in the pre-processing phase.

As the opinions are shown to be almost equally split regarding positive and negative sentiment, the positive tweets beside the slight numerical advantage also have a small negative difference in terms of the compound score, as shown in the figure below. Positive tweets have a mean of 0.525122, whereas the negative tweets a mean of -0.541433.

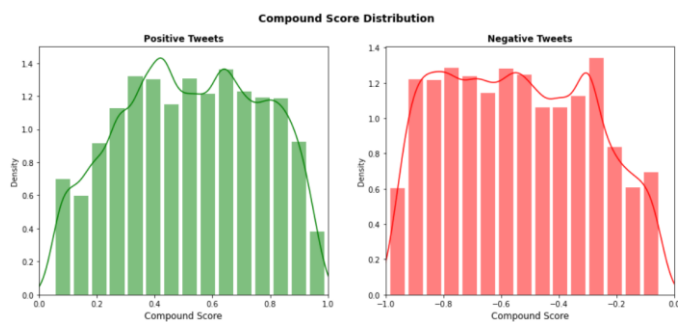


Fig4. Compound Score Distribution

Tweets were classified using the same method (VADER), and analysis resulted in 40.4% (1931) of tweets to be positive against 44.2% (2112) negative tweets, while 15.4% (735) are considered neutral.

Figure below illustrates the most frequent words on positive and negative tweets about Islam accordingly.

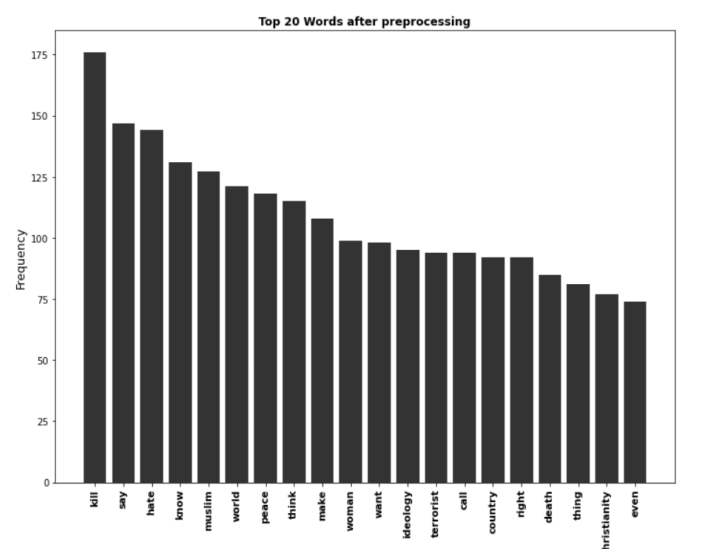


Fig5. Word Frequency based on Sentiment about Islam

3. Hate Speech Detection (SONAR)

This model is considered useful for our task, to investigate the presence of hate speech on tweets about religion, and to measure it. It will contribute to a general understanding of the religious landscape in an online environment. The process starts by loading the data in a dataframe format on the notebook. The data consist of about 48,500 tweets that are not pre-processed as the model is trained to process the input to extract the necessary features. Some pre-processing steps used in the model are: remove hashtags, remove links, remove mentions, tokenization, and stemming using Porter Stemmer.

SONAR Model

After the model is applied, the dataset consisting of 48,528 tweets is split into three categories **Hate Speech**, **Offensive Language**, and **Neither** (indicating neither hateful nor offensive). The first chart in figure below gives the overall results of tweets distribution across these categories. Hateful and offensive tweets combined resulted in a total of 3802 tweets or 7.83% of the dataset. Thus, leaving the non-problematic tweets to a total of 44726 or 92.16% of the dataset. Whereas in the second chart is shown the distribution among the so-called problematic tweets, hate speech results in a total of 232 Tweets (6.10% of problematic tweets) as Offensive Language dominates this category with a total of 3570 (93.89%).

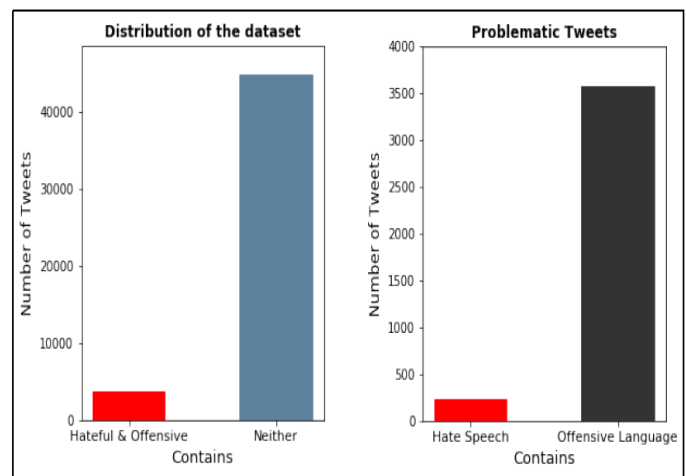


Fig6. Hate Speech Detection Results

It is interesting to see how the sentiment feature correlates with the hate speech classification results. Figure below shows the distribution of tweets per category of *hate speech*, *offensive language*, and *neither*, based on the sentiment, *being negative*, *positive*, or *neutral*. As expected, hate speech is occurred more in negative tweets, numerically speaking 60% (145 out of 232) tweets that contain hate speech are negative. In the case of offensive tweets, the proportions are similar to hate speech, as customarily expected the majority of tweets: 58.8% (2089 out of 3570 tweets) contain negative sentiment. From the rest, 1106 or 30.98% are positive, and 375 or 10.50% of tweets classified as offensive are neutral by sentiment.

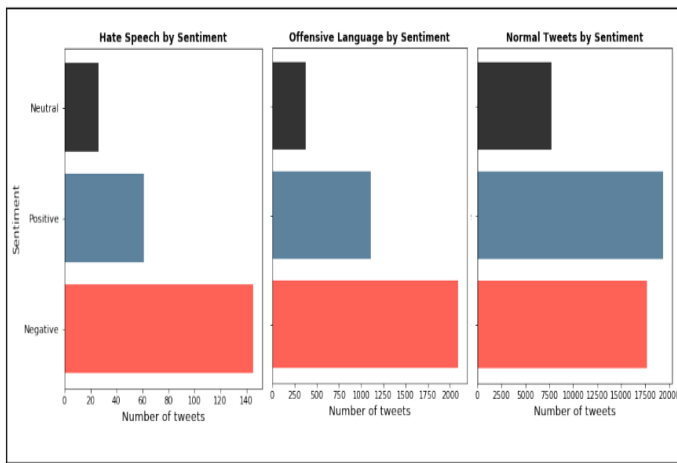


Fig7. Hate Speech Detection by sentiment

Figure below illustrates the distribution of hateful tweets based on their compound sentiment score. The left side corresponding to negative tweets is more populated (60% of tweets), which makes sense that a tweet using hate speech is negative in sentiment.

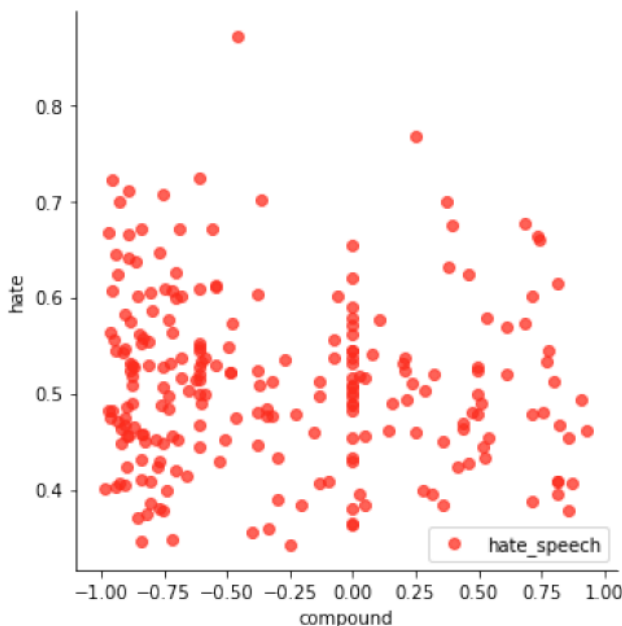


Fig8. Compound Sentiment Score

IV. RESULTS

LDA MODEL

After building the LDA model and running the optimal model we yield the following topics:

Topic	Topic_Keywords	Num_Tweets	Perc_Tweets	Text
0.0	0.0 peopl, muslim, reason, stop, tri, care, happen...	6877	14.52	[hope, last, littl, teach, group, faith, commu...
1.0	1.0 christian, freedom, state, teach, practic, gov...	6801	14.36	[page, today, essay, write, emeri, report, pro...
2.0	2.0 believ, true, wrong, call, fact, alway, real, ...	6241	13.18	[first, human, anim, appli, belief, magic, pow...
3.0	3.0 woman, world, follow, kill, church, control, e...	5955	12.58	[sikhism, respect, sidhu, respect, openli, app...
4.0	4.0 love, human, life, give, peac, live, respect, ...	5654	11.94	[certainli, give, context, anti, much, distrus...
5.0	5.0 race, polit, countri, hate, cultur, differ, ba...	5784	12.22	[cite, bibl, crime, minor, consid, hypocrisi, ...
6.0	6.0 make, thing, good, person, realli, problem, pe...	4865	10.28	[abraham, think, woman, less, right, problem, ...
7.0	7.0 faith, belief, believ, scienc, understand, fin...	5170	10.92	[scienc, educ, impact, christian, scientist, ...

Fig9. Topic with their keywords

Interpretation:

Topic [0] Religion & Politics is self-evidently represented by terms like politics, control, government, nation, trump, etc., which illustrate how religion is a sensitive aspect of its role related to politics.

Topic [1] Christianity is constituted with tweets which contain Christianity as the subject of the discussions and its representative key-terms such as church, catholic, christian etc., such topic is expected for an English-speaking audience mainly belonging to western world, having Christianity as the leading religion.

Topic [2] Religion & Science is concerned with the never-ending debate between religion and science, a debate intensified in the modern world which is led by a technological revolution.

Topic [3] Religious Doctrine, **[6] Personal Belief** and **[7] Diverse Opinions** seem to be closer to each other with slight differences, based on the keywords.

Topic [5] Islam is a collection of tweets with Islam as the main subject of discussion. As seen by the number of hashtags and the word-frequency statistics, Islam is highly discussed on social media for its different aspects, which trigger diverse reactions among people.

VADER MODEL

Figure below, gives the overall classification result of tweets to the sentiment class, where 42.2% (20524) of tweets are classified as *positive*. 41.1% (19988) were classified as *negative*, and the remaining 16.7% (8104) of tweets are considered *neutral* (neither positive or negative). The public opinion seems to be balanced in terms of the sentiments on discussions about religion on social media.

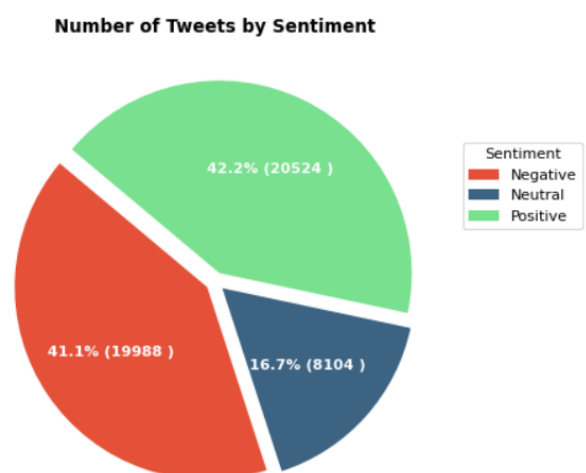


Fig10. Classification of Tweets by Sentiment

When it comes to specific religions, most of the discussions are oriented towards Islam with the highest number of tweets, 679 tweets after pre-processing steps, followed by Christianity with 127 tweets and then the other popular religions like Hinduism, Buddhism, and Judaism. **Islam** is the most discussed religion on

social media. Hence, we had a closer look into the sentiments towards Islam particularly. A sentiment analysis on tweets which talk about Islam yielded the following results:

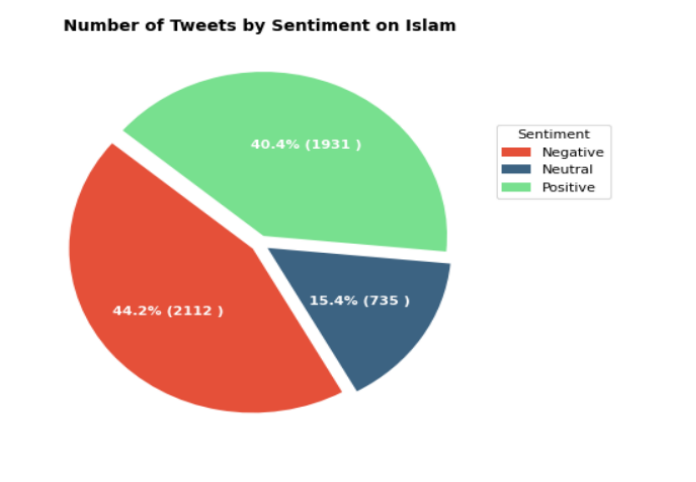


Fig11. Classification of Tweets by Sentiment on Islam

Top 10 most positive tweets (VADER MODEL):

tweet_text	tweet_text_p	negative	neutral	positive	compound	sentiment	lemmatized	no_stop_joined
0	Whn I was trying to focus more on DEEN someone...	0.000	0.818	0.382	0.9817	1	[try, focus, deen, tell, soon, feel, routine, ...]	try focus deen tell soon feel routine alhamdu...
1	# Islam is a beautiful # religion. It teaches patience...	0.000	0.528	0.472	0.9792	1	[beautiful, teach, love, importance, seek, kno...]	beautiful teach love importance seek knowledge...
2	Pronouncements in a religion does play its par...	0.000	0.631	0.369	0.9771	1	[pronouncement, play, part, take, partner, mak...]	pronouncement play part take partner make work...
3	FREEDOM TO CHOOSE RELIGION IS A HUMAN RIGHT!!	0.023	0.547	0.430	0.9738	1	[freedom, choose, human, right, freedom, wear...]	freedom choose human right freedom wear want w...
4	ADNAN OKTAR. For one thing, Islam is a of pea...	0.000	0.312	0.688	0.9688	1	[adnan, oktar, thing, affection, reward, smile, compassion, friends...]	adnan oktar thing affection compassion friends...
5	Islam is completely intolerant? Lmao a religio...	0.000	0.328	0.672	0.9663	1	[completely, intolerant, reward, smile, good, thought...]	completely intolerant reward smile good thought
6	allegation against Islam is that it is a relig...	0.000	0.662	0.338	0.9656	1	[oppress, woman, prophet, mohammad, also, attri...	oppress woman prophet mohammad also attribute ...
7	thats true, iran is a great country and deserv...	0.000	0.621	0.379	0.9639	1	[tr, true, great, country, deserve, glory, isla...]	s true great country deserve glory islam probl...
8	# WhatIKnowAboutIslam is that i am in love wit...	0.000	0.503	0.497	0.9628	1	[love, study, peace, harmony]	love study peace harmony
9	In every religion there is love e.g. be it Hinduism o...	0.026	0.608	0.365	0.9618	1	[love, culture, boundary, race, love, pure, be...]	love culture boundary race love pure beautiful...

Fig12. Top 10 most positive tweets (VADER Model)

The above-shown tweets are highly rated as positive, where the 10th tweet has a compound score of 0.9618, and the 1st tweet is rated with a compound score of 0.9817. The classifier has produced correct labels, as the tweets are clearly positive and do not hide underlying patters of irony or sarcasm that undermine the meaning of the sentences. Two tweet is about *Islam*, and one is about *Hinduism*. The rest are general opinions on religion focusing on the relationship with others, namely loving each other as the core religious principle one should follow. Most of the tweets (6/10) have a negative score of 0.000; other tweets have a very minimal above zero negative score because of the words: *intolerant*, *race*, *unsolicited*, etc. The tweet with the highest individual positive score is the 4th with a score of 0.688 because it mentions the word *peace* seven times, which increases the positivity of the sentence while it does not contain negation words.

Top 10 most negative tweets and their respective scores (VADER MODEL):

The above-shown tweets are very highly (in a negative sense) rated as negative where the 10th tweet has a compound score of -0.9758, and the 1st most negative tweet is rated with a compound score of -0.9878. It is evident that the classifier has produced correct labels, as the tweets are negative, expressing refusal, dislike, and negative feelings towards religion or a religious group i.e., Muslims. Some of the dominant words and their respective frequency on this table are death, Islam, rape etc. The most addressed religion is Islam, where majority of tweets focus on Islam when expressing their negative emotions, while one was about Christianity and another for the Buddhism/Hinduism in India.

tweet_text	tweet_text_p	negative	neutral	positive	compound	sentiment	lemmatized	no_stop_joined
0	That's what they live for. The desalination of...	0.649	0.351	0.000	-0.9878	-1	[live, desalination, other, hate, murder, rape...]	live desalination other hate murder rape pure ...
1	Threatening murder against a poor woman is not...	0.500	0.466	0.034	-0.9873	-1	[threaten, murder, poor, woman, protect, harm, enga...]	threaten murder poor woman protect harm engage...
2	Agreed, for all but Islam. It's a cult of war...	0.525	0.448	0.027	-0.9871	-1	[agree, death, call, confuse, public, behadin...]	agree death call confuse public behading rape...
3	Christ is dead too you know, but I hear y'all...	0.452	0.478	0.070	-0.9867	-1	[dead, know, hear, last, thing, need, bullshit...]	dead know hear last thing need bullshit push e...
4	That is not what China is saying. China is say...	0.417	0.532	0.050	-0.9847	-1	[say, say, anywhere, else, hate, enemy, enemy...]	say say anywhere else hate enemy enemy kiss fa...
5	This is exactly why we (USA) must declare Isla...	0.427	0.533	0.040	-0.9781	-1	[exactly, declare, country, ideology, make, ev...]	exactly declare country ideology make evil ev...
6	It's OK to track potential Islamic terrorists...	0.432	0.453	0.114	-0.9777	-1	[track, potential, islamic, terrorist, stop, d...]	track potential islamic terrorist stop domesti...
7	Murder is wrong in Islamic countries. Murder i...	0.442	0.505	0.053	-0.9773	-1	[murder, wrong, islamic, country, murder, wron...]	murder wrong islamic country murder wrong chri...
8	How is it a "thing"? Why is it okay to critici...	0.402	0.550	0.048	-0.9764	-1	[thing, criticise, idea, come, expose, alot, i...]	thing criticise idea come expose alot idea ins...
9	Disagree all u want, but u are wrong. He is le...	0.624	0.346	0.031	-0.9758	-1	[disagree, want, wrong, leader, deception, anti...]	disagree want wrong leader deception anti death

Fig13. Top 10 most negative tweets and their respective scores (VADER Model)

The above-shown tweets are very highly (in a negative sense) rated as negative where the 10th tweet has a compound score of -0.9758, and the 1st most negative tweet is rated with a compound score of -0.9878. It is evident that the classifier has produced correct labels, as the tweets are negative, expressing refusal, dislike, and negative feelings towards religion or a religious group i.e., Muslims. Some of the dominant words and their respective frequency on this table are death, Islam, rape etc. The most addressed religion is Islam, where majority of tweets focus on Islam when expressing their negative emotions, while one was about Christianity and another for the Buddhism/Hinduism in India.

SONAR MODEL

Top 10 most hateful tweets (SONAR MODEL):

tweet_text	negative	neutral	positive	compound	sentiment	Class	hate	offensive	neither
0	@ WritersOfColour "your religion is misogynist...	0.273	0.727	0.000	-0.4588	-1	hate_speech	0.870408	0.069669 0.059924
1	What the fuck is wrong with the human race? Re...	0.452	0.548	0.000	-0.7650	-1	hate_speech	0.824041	0.144219 0.031740
2	The systematic extermination of white race.wes...	0.000	0.863	0.137	0.4019	1	hate_speech	0.819088	0.136934 0.043978
3	What? White is a race. Judaism is a Religion s...	0.000	1.000	0.000	0.0000	0	hate_speech	0.813112	0.143532 0.043356
4	@ yesnicksearcy @ ksoderbl #AntiWhite anti-ra...	0.000	0.836	0.164	0.4486	1	hate_speech	0.809880	0.166541 0.023578
5	As a white ((Aryan)) who is the exact same rac...	0.089	0.911	0.000	-0.1901	-1	hate_speech	0.798556	0.130696 0.070748
6	White supremacy is an ideology. It has nothing...	0.000	0.870	0.130	0.1779	1	hate_speech	0.795342	0.161176 0.043482
7	So why that logic then it shouldn't be racist ...	0.104	0.896	0.000	-0.6124	-1	hate_speech	0.792905	0.170698 0.036397
8	@ BlushingGiraffe White is a race, anyone can ...	0.156	0.596	0.248	0.2509	1	hate_speech	0.767149	0.061211 0.171641
9	Using one person/ persons to make a negative s...	0.214	0.677	0.109	-0.7163	-1	hate_speech	0.759593	0.180686 0.056721

Fig14. Top 10 most hateful tweets

Out of 232 tweets classified as hate speech, the above-presented tweets have the highest hate score, the 10th tweet has a hate score of 0.759593, and the 1st tweet is rated with a hate score of 0.870408.From a general manual analysis, we see that these tweets are dominated by a presence of offensive words like tweet number. The model is trained to overcome the thin border between offensive language and hate speech by focusing on a vocabulary of hateful words. The most dominant aspect of hate is race. Eight out of 10 most hateful tweets contain a reference to white people, using words like *white feminist*, *white racist Americans*, *white folks*, *white Nationalism*, *white European descent*, *white Men*, and *white boys club*. Tweet number expresses hate against Muslims calling their religion *dated*, *sexist*, *stupid*, while tweets number and contain references to Jews. Tweet number is hated speech with a racist motive with reference to black people using the slang words.

Top 10 most Offensive tweets (SONAR MODEL):

	tweet_text	negative	neutral	positive	compound	sentiment	Class	hate	offensive	neither
0	dustin speaking "son of a bitch" is my religion	0.388	0.612	0.000	-0.5859	-1	offensive_language	0.010516	0.989317	0.000167
1	Anja Nissen saying "vegan bitches!" in her las...	0.000	1.000	0.000	0.0000	0	offensive_language	0.009915	0.989315	0.000770
2	Spencer Reid saying "son of a bitch" is my rel...	0.352	0.648	0.000	-0.5859	-1	offensive_language	0.010632	0.988830	0.000538
3	No more drugs for me, pussy n religion is all ...	0.216	0.784	0.000	-0.2960	-1	offensive_language	0.011703	0.987489	0.000808
4	Honestly tho. Pussy and religion is all I need	0.000	0.667	0.333	0.4588	1	offensive_language	0.012881	0.986379	0.000740
5	No more drugs for me, pussy & religion is all ...	0.216	0.784	0.000	-0.2960	-1	offensive_language	0.012452	0.986363	0.001185
6	no more drugs for me pussy & religion is all i...	0.216	0.784	0.000	-0.2960	-1	offensive_language	0.012452	0.986363	0.001185
7	No more drugs for me, pussy & religion is all ...	0.216	0.784	0.000	-0.2960	-1	offensive_language	0.012452	0.986363	0.001185
8	No more drugs for me pussy & religion is all i...	0.216	0.784	0.000	-0.2960	-1	offensive_language	0.012452	0.986363	0.001185
9	RT @ TeamKanyeDaily: No more drugs for me, pus...	0.180	0.820	0.000	-0.2960	-1	offensive_language	0.012096	0.986205	0.001729

Fig15. Top 10 most offensive tweets

Offensive language scores dominate the ratings of the list as it varies from 98.6% to 98.9% for the first tweet. Only tweet number [10] has references to Islam, and tweet number [5] uses the word Hebrew but with no clear evidence of any reference to Judaism. Offensive tweets are populated vastly by offensive words of sexual reference such as *bitch*, which appears in 8 out of 10 tweets of the list. Tweets [3, 5, 6] contain strong sex-related words, while only tweets [1, 2] use the phrase "is my religion" in a figurative way.

V. CONCLUSION

As we can see from above results, 42.2% (20524) tweets are positive tweets on religion, while 41.1% (19988) tweets are negative tweets while other are neutral. Based on this, we can see that no. of positive and negative tweets are almost same. This shows that people don't think that they are hurting someone sentiment.

Data analysis on tweets on Islam shows that people usually have negative sentiment towards it and associate negative words with Islam. The results show 44.2% negative tweets while only 40.4% positive tweets (other being neutral). Most common negative word is "Kill" which occurs 176 times in tweets on Islam.

When analysing hate speech detection result, we see that only 0.5% of all tweets are hateful and 7.4% are offensive while a staggering 92.1% tweets are neutral in this case. This can be further used in real time to detect hateful and offensive tweets which can be promptly deleted to avoid inciting a particular ethnic group(s). This will lead to avoid chaos.

VI. FUTURE SCOPE

This project method can be later generalized and used in finding negative tweets in real time which can be later removed to prevent igniting a certain ethnic group and can be used in avoiding chaos.

VII. ACKNOWLEDGEMENT

This research was supported by Vishwakarma Institute of Technology. We thank Prof Rupali Tornekar for assistance with Machine Learning and its various algorithms. We would also like to show our gratitude to the Prof Rupali Tornekar for sharing their pearls of wisdom with us during the course of this research.

VIII. REFERENCES

- [1] J. Wiebe and E. Riloff. Creating subjective and objective sentence classifiers from unannotated texts. In In CICLing2005, pages 486– 4972005.
- [2] Hutto, C. J., & Gilbert, E. (2014). VADER: A Parsimonious Rule-based Model for Sentiment Analysis of social media Text. *Eighth international AAAI conference on weblogs and*

social media. Michigan: AAAI Press.

[3] Davidson, T., Warmesley, D., Macy, M., & Weber, I. (2017). Automated Hate Speech Detection and the Problem of Offensive Language. Eleventh International AAAI Conference on Web and social media (pp. 512–515). Montreal: AAAI Press.

[4] K. Dave, S. Lawrence, and D. Pennock. Opinion extraction and semantic classification of product reviews. In Proceedings of the 12th International World Wide Web Conference (WWW), pages 519–528, 20

[5] Blei, D. M., Ng, A. Y., & Jordan, M. I. (2003, 01). Latent Dirichlet Allocation. (J. Lafferty, Ed.) *Journal of Machine Learning Research*, 993–1022.

